

# DB Connection with Python

## 1. Required Libraries

```
pip install pandas sqlalchemy pymysql mysql-connector-python python-dotenv
```

## 2. Database Credentials (.env file)

```
DB_USER=root
DB_PASSWORD=your_password
DB_HOST=localhost
DB_PORT=3306
DB_NAME=ship_db
```

## 3. Load Environment Variables

```
from dotenv import load_dotenv
import os

load_dotenv()

DB_USER = os.getenv("DB_USER")
DB_PASSWORD = os.getenv("DB_PASSWORD")
DB_HOST = os.getenv("DB_HOST")
DB_PORT = os.getenv("DB_PORT")
DB_NAME = os.getenv("DB_NAME")
```

## 4. Create MySQL Connection using SQLAlchemy

```
from sqlalchemy import create_engine

engine = create_engine(
    f"mysql+pymysql://{DB_USER}:{DB_PASSWORD}@{DB_HOST}:{DB_PORT}/{DB_NAME}"
)
```

## 5. Read Titanic Table into Pandas

```
import pandas as pd

query = "SELECT * FROM titanic"
df = pd.read_sql(query, engine)

df.head()
```

## 6. Role of Each Library (Quick Reference)

|                 |                                |
|-----------------|--------------------------------|
| SQLAlchemy      | → Connection manager           |
| PyMySQL         | → MySQL driver                 |
| dotenv          | → Credential security          |
| mysql-connector | → Official low-level connector |

These libraries work together to enable secure, reliable, and efficient database connectivity between Python and MySQL for data analysis workflows.